

Entanglement Swapping in Quantum Repeater Chains

Monte Carlo Simulation with QuantumSavory.jl

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Outline

Theory

1. Motivation and quantum repeaters
2. Entanglement swapping protocol
3. Noise model: depolarization

Simulation & Results

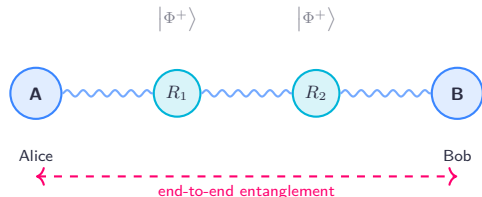
4. Simulator architecture
5. Ideal case validation
6. Monte Carlo simulation
7. Numerical vs. theoretical comparison
8. Emergent effects and conclusions

Why Quantum Repeaters?

- ▶ Optical fiber: **exponential losses** with distance
- ▶ **No-cloning theorem**: no classical amplification
- ▶ **Quantum repeaters**: hop-by-hop entanglement distribution

Project Goal

Simulate Alice $\leftrightarrow$ Bob entanglement across N repeaters, measuring **fidelity** and **distribution time** vs noise.



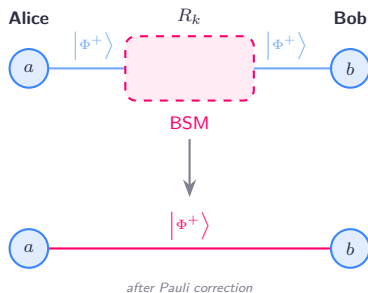
The Protocol: Entanglement Swapping

Key idea: a BSM at the repeater teleports entanglement to the remote nodes.

For each repeater R_k :

1. Holds two qubits entangled with left and right nodes
2. **BSM** on the two local qubits
3. Outcome determines a Pauli correction
4. Remote nodes end up entangled in $|\Phi^+\rangle$

N repeaters $\rightarrow N$ sequential swaps $\Rightarrow$ Alice and Bob share $|\Phi^+\rangle$.



Noise Model: Depolarization

Quantum memories **depolarize** while waiting.

Single-qubit depolarizing channel
(QuantumSavory convention):

$$\mathcal{E}(\rho) = (1 - p) \rho + p \frac{I}{2}$$

Noise tied to waiting time:

$$p(\Delta t) = 1 - e^{-\Delta t/\tau}$$

$\tau = -1/\ln(1 - p_w)$, p_w = per-step depolarization rate.

Fidelity: $F = \frac{1}{4}(1 + \langle XX \rangle + \langle ZZ \rangle - \langle YY \rangle)$

Why Time Matters

If $p_s < 1$, faster links **wait** for slower ones.

$$T = \max_{i=1}^{N+1} T_i, \quad T_i \sim \text{Geom}(p_s)$$

Larger $T \Rightarrow$ more depolarization $\Rightarrow$ lower F .

Assumption: all BSMs fire at T , classical messages instantaneous. Swapping *as soon as possible* would free repeater qubits earlier $\Rightarrow$ synchronous model is the pessimistic case.

Simulator Architecture

Language: Julia **Framework:**
QuantumSavory.jl



Linear network: Alice + N repeaters + Bob

- ▶ Alice and Bob: 1 memory slot each
- ▶ Each repeater: 2 slots (left + right)
- ▶ Topology: linear graph (RegisterNet)

Simulation Pipeline

1. Bell pair generation on each link
2. Time evolution (uptotime!) $\rightarrow$ depolarization
3. Sequential swapping: N BSMs
4. Fidelity measurement + MC (M runs)

Ideal Case Validation

Test: no noise ($p_w = 0$), deterministic ($p_s = 1$) $\Rightarrow$ fidelity must be **exactly 1.0**.

N	F	Status
1	1.000	OK
2	1.000	OK
3	1.000	OK
5	1.000	OK

Swapping preserves $|\Phi^+\rangle$ perfectly without noise.

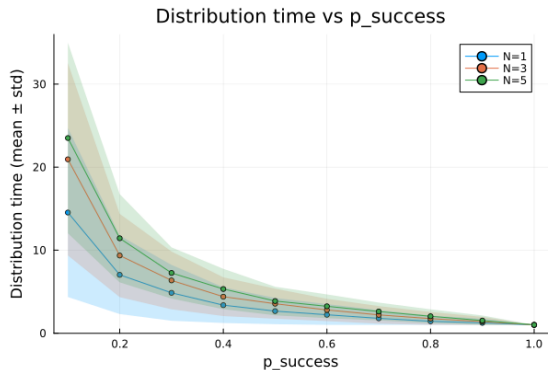
What it Validates

- ▶ Bell pair initialization
- ▶ BSM + Pauli corrections
- ▶ Sequential swap composition ($\forall N$)
- ▶ Fidelity via $\langle XX \rangle, \langle ZZ \rangle, \langle YY \rangle$

Entry point

```
julia -project=. run_ideal.jl
```

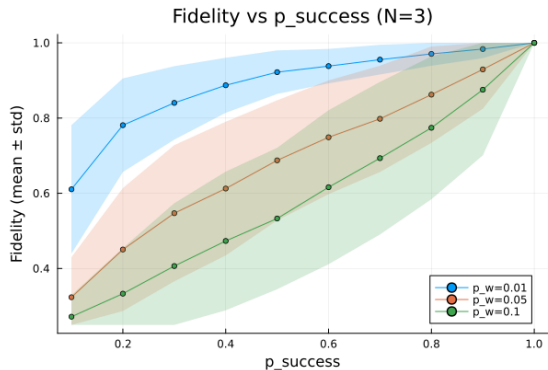
MC Simulation: Distribution Time



Parameters: $p_w = 0$, $M = 500$ runs

- ▶ Time scales as $\sim 1/p_s$
(geometric attempts per link)
- ▶ Bottleneck: **slowest link**
 $T = \max_i T_i$
- ▶ More repeaters $\Rightarrow$ more links $\Rightarrow$ larger max
- ▶ Bands: ± 1 std of the per-run distribution, clipped to $T \geq 1$
(high spread at low p_s)

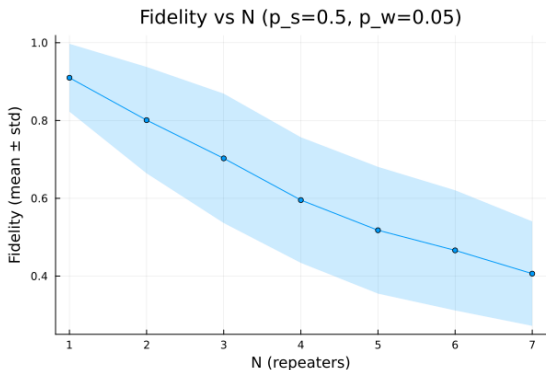
MC Simulation: Fidelity vs p_s



Parameters: $N = 3$, $M = 500$ runs

- ▶ Low $p_s \rightarrow$ long wait $\rightarrow$ more depolarization $\rightarrow F$ drops
- ▶ $p_w = 0.10$ degrades far more than $p_w = 0.01$
- ▶ At $p_s = 1$: $T = 1$ and no inter-link waiting, so $F \rightarrow 1$
- ▶ Bands: ± 1 std, clipped to the physical range $[0.25, 1]$
- ▶ **Tradeoff:** reliable links reduce both time and degradation

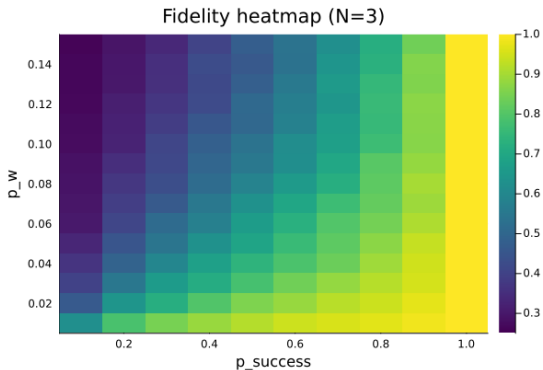
MC Simulation: Fidelity vs N



Parameters: $p_s = 0.5$, $p_w = 0.05$,
 $M = 500$

- ▶ F decreases monotonically with N
- ▶ Two effects: more noisy qubits + larger $T = \max_i T_i$
- ▶ $F \approx 0.91$ ($N=1$) to $F \approx 0.41$ ($N=7$)
- ▶ Band (± 1 std) widens with N : more uncertainty

MC Simulation: Fidelity Heatmap



Parameters: $N = 3$, $M = 500$

2D mean fidelity in (p_s, p_w) space.

- ▶ **Bottom-right** (yellow): high p_s , low $p_w \Rightarrow F \rightarrow 1$
- ▶ **Top-left** (purple): low p_s , high $p_w \Rightarrow F \rightarrow 0.25$ (maximally mixed)
- ▶ Gradient shows **non-linearity**: at low p_s the waiting time amplifies memory noise
- ▶ Right edge: $F = 1$ for every p_w (zero waiting time)

Theoretical Framework

Distribution Time

For $N + 1$ i.i.d. links with $T_i \sim \text{Geom}(p_s)$:

$$\mathbb{E}[T] = \sum_{t=0}^{\infty} \left[1 - (1 - (1 - p_s)^t)^{N+1} \right]$$

Harmonic approximation:

$$\mathbb{E}[T] \approx \frac{H(N+1)}{p_s}, \quad H(n) = \sum_{k=1}^n \frac{1}{k}$$

Werner Model for Fidelity

Each link i waits $\Delta t_i = T - T_i$ time steps.
Werner parameter per link:

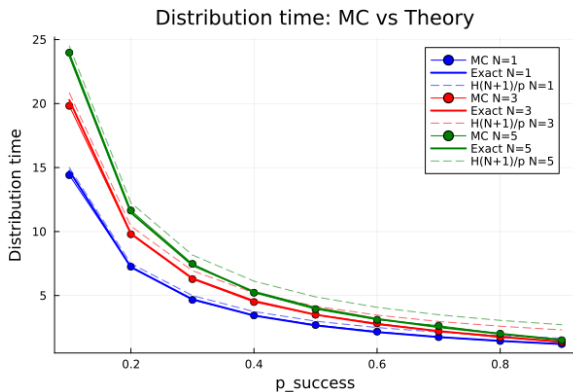
$$w_i = (1 - p(\Delta t_i))^2$$

where $p(\Delta t) = 1 - (1 - p_w)^{\Delta t}$. Swapping multiplies Werner parameters:

$$w_{\text{tot}} = \prod_{i=1}^{N+1} w_i \quad F = \frac{1 + 3 w_{\text{tot}}}{4}$$

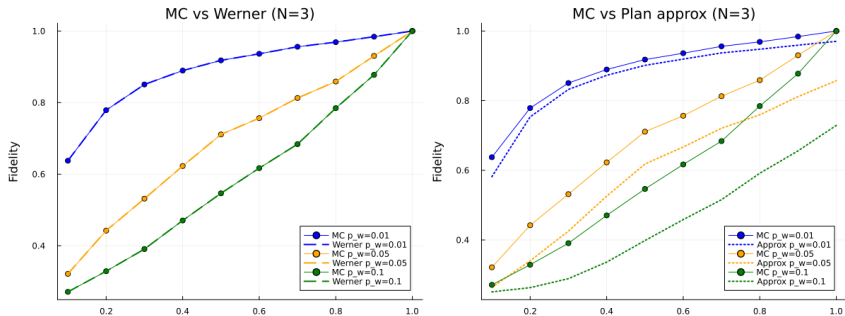
Exact for ideal BSMs $\Rightarrow$ closed-form prediction.

Analysis: MC Time vs Theory



- ▶ MC data closely follow **exact curves** (errors $< 5\%$, $M = 1000$)
- ▶ Harmonic approx. $H(N+1)/p_s$ accurate at **low** p_s (geometric $\rightarrow$ exponential limit), overestimates at high p_s
- ▶ Exact formula captures order statistics of i.i.d. geometric RVs
- ▶ Log growth with N : see appendix

Analysis: MC Fidelity vs Werner vs Approximation

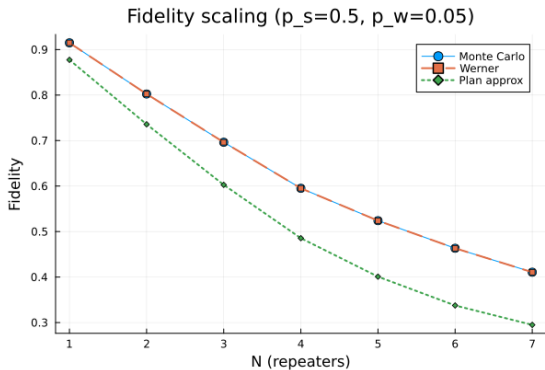


Left panel: per-run *paired* comparison (same generation times): Werner prediction and simulation coincide to **machine precision** ($\sim 10^{-16}$) — a closed form, not a fit.

Right panel: $F \approx (1 - p_w)^{TN}$ underestimates (heuristic from the project plan, consistently pessimistic).

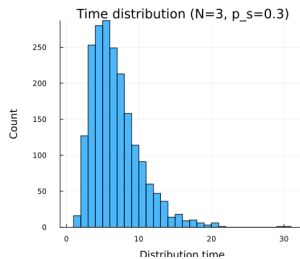
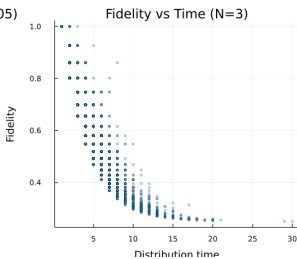
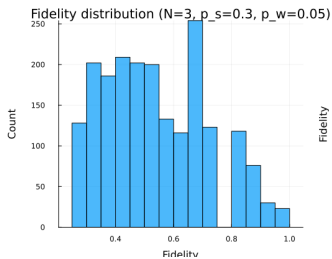
Analysis: Fidelity Scaling with N

Parameters: $p_s = 0.5$, $p_w = 0.05$



- ▶ Werner curve **coincides** with MC at every N (per-run paired, machine precision)
- ▶ Why exact: depolarized pairs *are* Werner states, and ideal BSMs multiply w exactly
- ▶ Plan approx. diverges at large N (too pessimistic)
- ▶ **Exponential decay** in N : each repeater multiplies the Werner parameter

Emergent Effects



Distribution of F : broad, left-skewed mass down to 0.25 (mixed state); the spikes are the *discrete* values of F .

F vs T : clear **negative correlation**; horizontal bands: F is set by the integer waits $T - T_i$, so it is *discrete* and still varies at fixed T .

Distribution of T : right-skewed with a geometric (exponential) tail, outliers wait much longer.

Conclusions

Main Results

- ▶ Swapping preserves $|\Phi^+\rangle$ perfectly (**validated** $N = 1 \dots 5$)
- ▶ F degrades with N , p_w , low p_s (noise + waiting time)
- ▶ **Werner** closed form: **exact** agreement with MC (machine precision)
- ▶ **Plan approximation**: heuristic, consistently pessimistic

Future Work

- ▶ Entanglement purification to improve F
- ▶ Adaptive generation protocols (non-i.i.d.)
- ▶ Non-linear topologies (trees, graphs)
- ▶ More realistic noise models (dephasing, losses)

Code

```
run_ideal.jl | run_simulation.jl |  
run_analysis.jl
```

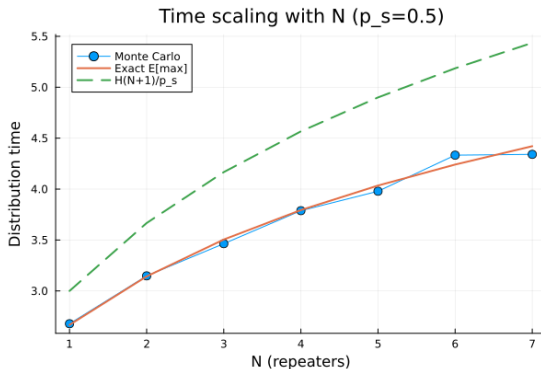
Thank You

Questions?

Appendix: Simulation Parameters

Parameter	Values	Description
N	1, 3, 5 (sim.) / 1–7 (analysis)	Number of repeaters in the chain
p_s	0.1–1.0, step 0.1	Bell pair generation success probability
p_w	0.01, 0.05, 0.10	Per-time-step depolarization parameter
M	500 (sim.) / 1000–2000 (analysis)	Number of Monte Carlo iterations
τ	$-1/\ln(1 - p_w)$	Depolarization time (QuantumSavory)
Seed	2025	Global RNG seed (reproducible figures)

Appendix: Time Scaling with N



Parameters: $p_s = 0.5$, $p_w = 0$,
 $M = 1000$

- ▶ $\mathbb{E}[T] = \mathbb{E}[\max_{i \leq N+1} T_i]$ grows **logarithmically** in N
- ▶ Harmonic approximation $H(N+1)/p_s$ overestimates slightly at moderate p_s
- ▶ Doubling the chain adds little latency — waiting is dominated by $1/p_s$